② Creates a security group used to grant network traffic from the EC2 instance to EFS

③ Allows incoming traffic on port 2049 used by NFS

④ The mount target provides a network interface for the filesystem.

⑤ The mount target is attached to a subnet.

⑥ Therefore, you need a mount target per subnet.

⑦ The blueprint used by the Auto Scaling group to launch virtual machines

⑧ Creates a folder used by Jenkins to store data if it does not exist yet

⑨ Adds an entry to the configuration file for volumes

⑩ Waits until the EFS filesystem becomes available

Mounts the EFS filesystem

Changes the ownership of the mounted directory to make sure Jenkins is able to write and read files

Creates the Auto Scaling group

References the launch template defined above

You can find the CloudFormation template for this example on GitHub and on S3. Download a snapshot of the repository at https://github.com/AWSinAction/code3/archive/main.zip. The file we're talking about is located at chapter13/multiaz-efs .yaml. On S3, the same file is located at https://s3.amazonaws.com/awsinaction-code3/chapter13/multiaz-efs.yaml.

Execute the following command to create the new Jenkins setup that stores state on EFS. Replace `$Password` with a password consisting of 8–40 characters:

```
$ aws cloudformation create-stack --stack-name jenkins-multiaz-efs \
➥ --template-url https:/ /s3.amazonaws.com/\
➥ awsinaction-code3/chapter13/multiaz-efs.yaml \
➥ --parameters "ParameterKey=JenkinsAdminPassword,
➥ ParameterValue=$Password" \
➥ --capabilities CAPABILITY_IAM
```

The creation of the CloudFormation stack will take a few minutes. Run the following command to get the public IP address of the virtual machine. If no IP address appears, the virtual machine isn't started yet. In this case, wait another minute, and try again:

```
$ aws ec2 describe-instances --filters "Name=tag:Name,\
➥ Values=jenkins-multiaz-efs" "Name=instance-state-code,Values=16" \
➥ --query "Reservations[0].Instances[0].\
➥ [InstanceId, PublicIpAddress, PrivateIpAddress, SubnetId]"
[
  "i-0efcd2f01a3e3af1d",    ①
```